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Sequential Determination of Serum Human Placental Lactogen, Estriol, and Estetrol for Assessment of Fetal Morbidity

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Serial measurements were made of the concentrations of maternal serum human placental lactogen (hPL) (300 determinations), estriol (E_3) (460 determinations), and estetrol (E_4) (275 determinations) in normal human pregnancy during the third trimester period. Simultaneous determinations of serum hPL, E_3, and E_4 were also made sequentially on blood samples from 6 diabetic and 5 toxemic pregnant women to ascertain the relative usefulness of these parameters as indicators of fetal welfare. In uncomplicated diabetic patients controlled with insulin, all parameters increased with gestational age. In three pregnancies complicated by severe toxemia in which fetal distress progressed to intrauterine fetal death, both serum E_3 and E_4 levels decreased progressively, but E_4 concentration started to decrease at least 1 day earlier than E_3 prior to fetal death. In other women, the E_4 levels appeared to drop or decrease significantly whereas the E_3 levels remained almost unchanged. Daily hPL levels remained low in chronic fetal distress and, therefore, appeared to be of minimal value for predicting either intrauterine death or acute fetal distress. Therefore, serum E_4 measurement seems to provide a more sensitive and reliable indicator of fetal morbidity than the measurement of serum E_3 during toxemic pregnancies.

THE DAILY EXCRETION of estriol (E_3) in urine and the concentration of E_3 and human placental lactogen (hPL) in plasma are considered to be indices of fetal well-being in late pregnancy.[1–4] Studies on the measurement of urinary E_3,[5] plasma estrogens,[6–10] and most recently, plasma estradiol (E_2) and estetrol (E_4)[11] after dehydroepiandrosterone sulfate (DHEA-sulfate) loading, are inconclusive and difficult to interpret.

Obstetricians need more reliable tools with which to assess fetal morbidity. One-half of all perinatal deaths occur prior to labor and delivery. The precise predictability of intrauterine fetal well-being, particularly in high-risk pregnancies, is still lacking. No single laboratory test has a proven predictive value for assessing the complex fetal–maternal–placental function.

Estetrol (E_4), isolated from the urine of newborns and pregnant women, is a characteristic product of 15α-hydroxylation of its precursor in fetal liver, and has been suggested as a good indicator of fetal health. The measurement of serum E_4 by radioimmunoassay (RIA) has been delayed because ^{3}H-E_4 with high specific activity and suitable antisera were not available. Development of RIA methods for plasma E_4 is more recent. Gurpide et al[4] measured E_4 using a sheep antiestriol-16,17-dihemisuccinate-BSA antiserum which had high cross-reactivity with E_4. Using antisera developed by Fishman and Guzik,[12] Korda et al[13] determined the plasma E_4 values during the last half of normal pregnancy, and Tulchinsky et al[14] were able to measure maternal plasma E_4 levels of several abnormal pregnancies, such as un-

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complicated diabetic, severe preeclamptic, anencephalic, and severe Rh-isoimmune sensitized patients. It has been suggested that E_4 measurements provide the obstetrician with approximately the same type of information received from E_3 determination. Recently, Kundu and Grant,[15] in our laboratory, reported the production of a highly specific antiserum against E_4, its characteristics, and a RIA method for the measurement of E_4 in the sera of normal women in their third trimester of pregnancy. Two more antisera raised against E_4 have also been reported[16,17] recently, but no methods have been described to determine E_4 levels from clinical samples.

In order to compare directly the relative usefulness of the maternal serum E_4 concentration with those of two other parameters, viz., serum E_3 and hPL, at first a normal distribution of each parameter was determined by analyzing 275 (for E_4), 460 (for E_3), and 300 (for hPL) clinically normal pregnancy sera during the third trimester of pregnancy. The mean curve and limits of normal variation of each parameter have also been established. Having established the normal values of each parameter of each week of gestation, simultaneous measurement of serum E_4 (with the rate limiting step in synthesis being 15α-hydroxylation by the fetal liver), E_3, and hPL levels (predicting placental mass and function) were performed sequentially on several complicated high-risk patients. Most high-risk pregnancies were managed according to the clinical course and the results of serum E_3 and hPL determinations and oxytocin challenge tests; serum E_4 data were analyzed retrospectively. The results of this study are the subject matter of this report.

MATERIALS AND METHODS

Measurement of Serum E_4

Serum E_4 was measured by a RIA method using a highly specific antisera raised in rabbits against a new E_4 antigen, estetrol-6-carboxymethyloximino-bovine serum albumin. The characteristics of the antisera and the usual parameters (specificity, accuracy, sensitivity, and precision) of the RIA method are described in detail in a previous publication.[15] Each serum sample was analyzed in duplicate along with a quality control. Whenever an abnormal high or low value is obtained, a repeat analysis from the same serum is performed to establish its reliability.

Measurement of Serum E_3

Serum E_3 was determined by a RIA method using a commercial kit.* The antiserum used was prepared in rabbits against E_3-6-conjugate which was considered by several investigators to be the most specific for E_3. Each sample was analyzed in duplicate along with quality control.

Measurement of Serum hPL

This was also determined by a RIA method using a commercial hPL-RIA kit† Each sample was analyzed in duplicate along with quality control.

Monitoring of the Fetal Heart Rate

Antenatal fetal heart rate monitoring of short term beat-to-beat variability and periodic heart rate change during the oxytocin challenge test (OCT) was determined with a Mennan-Greatbatch research fetal monitor. Oxytocin challenge tests were interpreted by Freeman's criteria.[18]

Apgar Score

This was evaluated at 1 and 5 minutes after birth.

Human Subjects

Normal pregnant women who regularly visited the Women's Clinic, operated by the Children's Hospital of

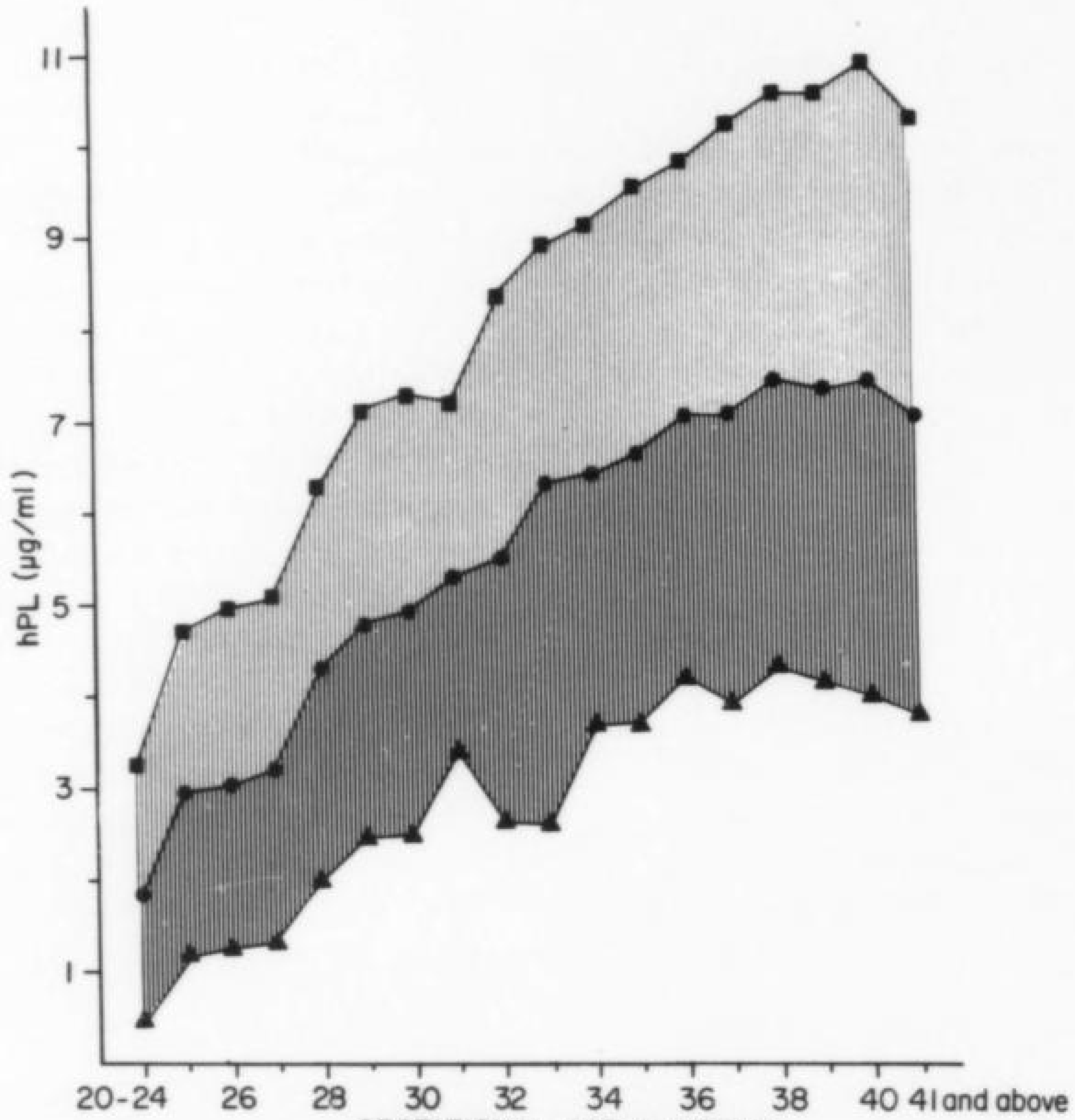

Fig 1. Distribution of serum hPL levels from normal pregnant women during the third trimester of their pregnancies. The middle curve (⊙—⊙—⊙) represents the mean value ± 2 SD. The upper (▣—▣—▣) and lower (△—△—△) curves represent the upper and lower 95% confidence limits. Three hundred values were used with a minimum 5 and a maximum 40 samples in a particular week.

* New England Nuclear Corp., Boston, Mass.

† Amersham-Searle, Arlington, Ill.

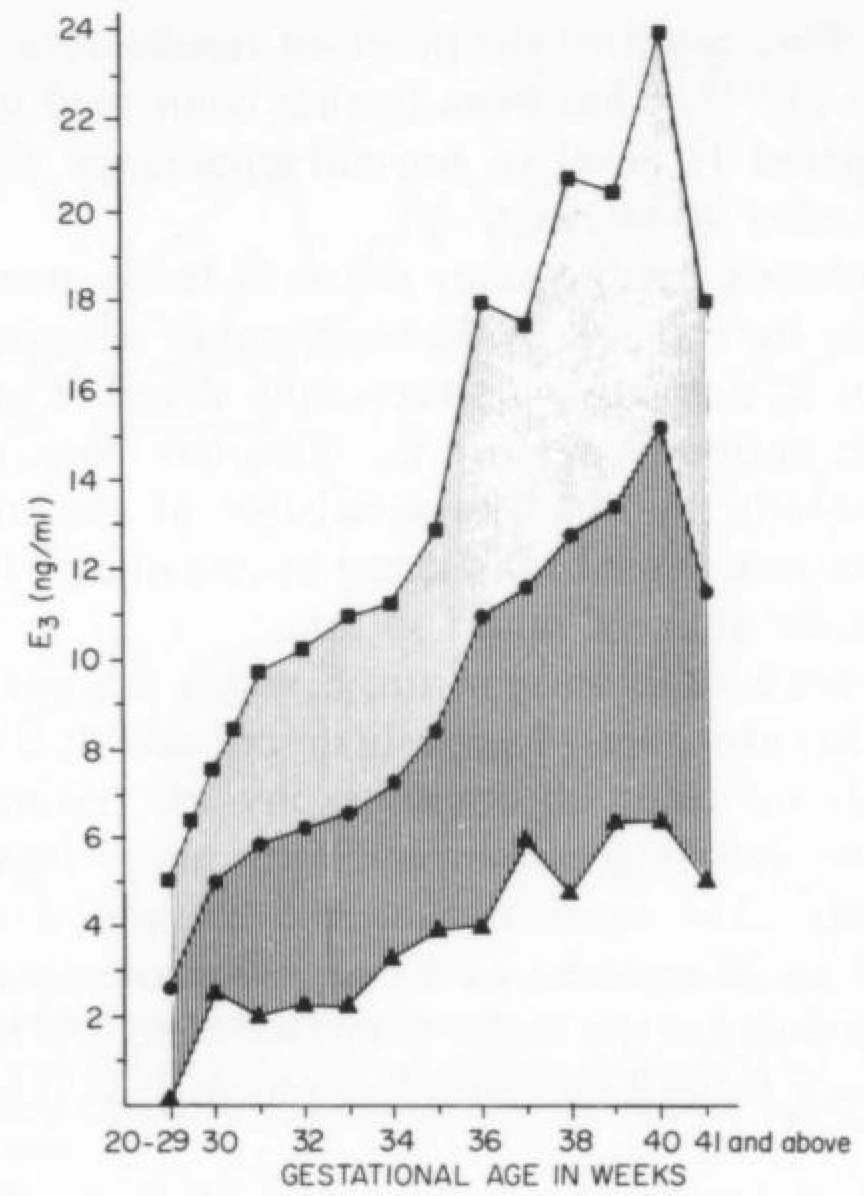

Fig 2. Distribution of serum unconjugated E_3 levels from normal pregnant women during the third trimester of their pregnancies. The middle curve (⊙—⊙—⊙) represents the mean value ± 2 SD. The upper (□—□—□) and lower (△—△—△) curves represent the upper and lower 95% confidence limits. Four hundred sixty values were used with a minimum 15 and a maximum 90 samples in a particular week.

Buffalo, Indian Health Services at Rosebud, Wagner, Eagle Butte, Chamberlain, and Pine Ridge Hospitals, and Yankton Clinic and Medical Clinic of Yankton, were selected for the purpose of establishing the normal levels of maternal serum E_4, E_3, and hPL. Their gestational age was recorded from their last menstrual period, confirmed by the date of birth of their babies. Eleven pregnant subjects with complications of pregnancy, and who were hospitalized either in Children's Hospital of Buffalo or Sacred Heart Hospital in Yankton, were chosen for study. Among them, 2 had uncomplicated diabetes, 1 had diabetes complicated with urinary tract infection, 4 had diabetes complicated with superimposed preeclampsia, and 4 had toxemia without diabetes. All signed an informed consent form approved by the Review Committee on Investigations Involving Human Subjects at the Children's Hospital of Buffalo and the University of South Dakota Medical School. All patients had their gestational ages estimated by obstetric examination and serial sonography.

Serum Samples

During the scheduled visits to the clinic as a routine check-up, venous blood samples (about 10 ml) were collected from the normal pregnant women. Similarly, blood was drawn from hospitalized patients almost daily in the morning. After clotting in the cold, these were centrifuged at 3000 rpm in the cold, and the clear sera were frozen and sent to our laboratory in frozen state. They were stored frozen until assayed. Serum E_4, E_3, and hPL determinations were performed according to the previously described procedure[15] or the protocol supplied by commercial sources.

RESULTS

Figure 1 shows the normal distribution of hPL according to the gestational age of the pregnancy. The mean value lies between 2 and 7 μg/ml depending on the duration of pregnancy which is in good agreement with other published values.[19] The value normally increases as pregnancy duration progresses until shortly before term when the concentration remains stable or, in some cases, decreases.

Figure 2 similarly shows the normal distribution of E_3. The mean value lies between 4 and 15 ng/ml depending on the gestational age which is similar to those obtained by other investigators.[20-22]

Figure 3 represents the normal distribution of E_4 in each week at the third trimester of pregnancies. The mean value lies between 250 and 950 pg/ml of serum depending on gestational age.

Having established the 95% confidence limits of serum hPL, E_3, and E_4 in the normal pregnant population, selected normal pregnant subjects were chosen for the analysis of the above parameters sequentially. A total of 40 normal patients have been analyzed individually and some of the data are given in Figure 4. It is clear that as pregnancy progresses, all the values rise until the 40th week; after that all the values tend to decrease. This is observed both in group and individual analysis.

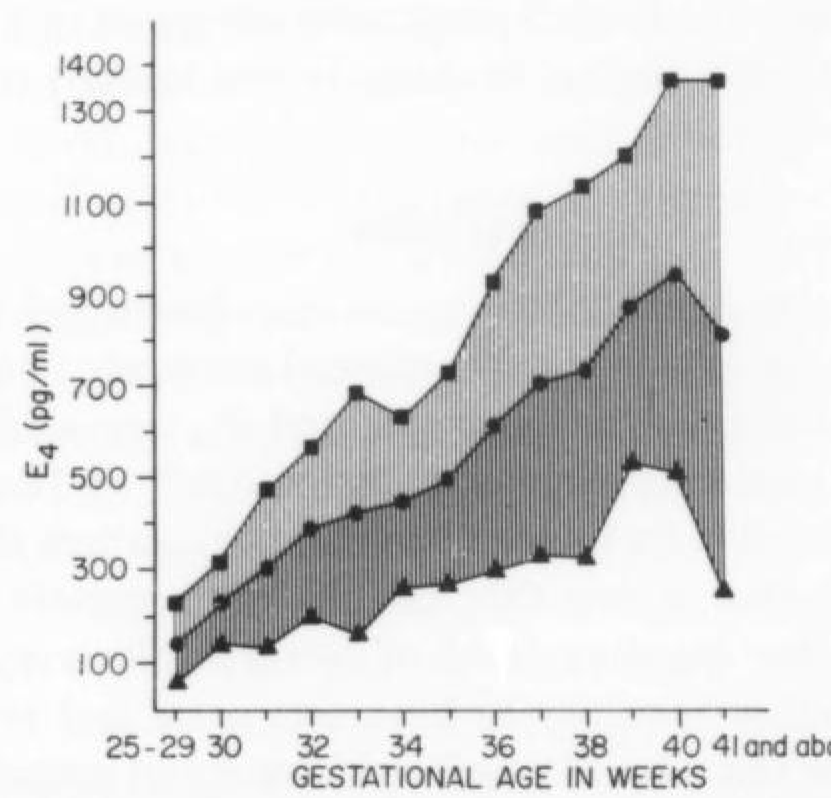

Fig 3. Distribution of serum unconjugated E_4 levels from normal pregnant women during the same period as above. The middle curve (⊙—⊙—⊙) represents the mean value ± 2 SD. The upper (□—□—□) and lower (△—△—△) curves represent the upper and lower 95% confidence limits. Two hundred seventy-five values were used with a minimum 12 and a maximum 35 samples in a particular week.

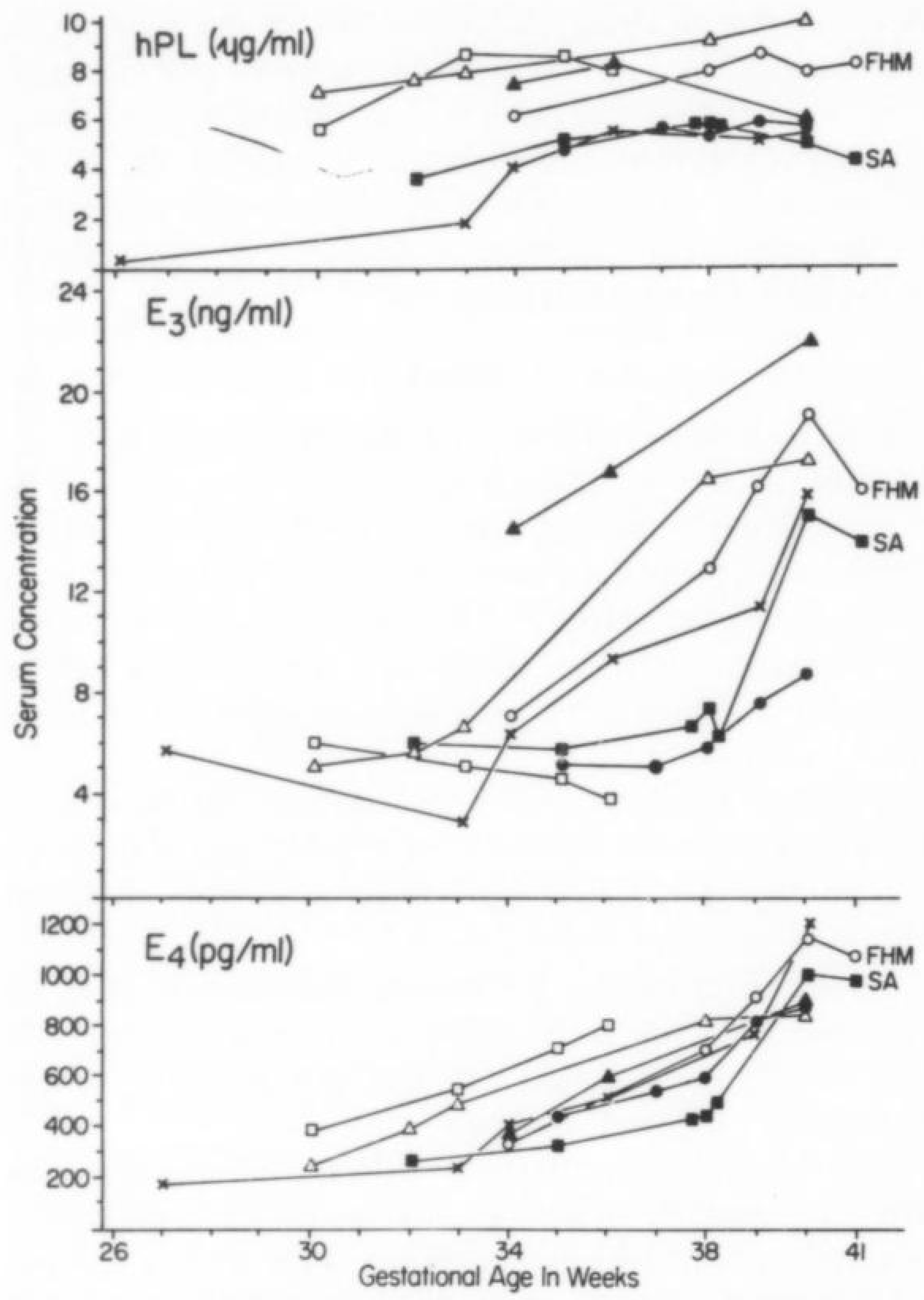

Fig 4. Serial serum E_4, E_3, and hPL in 7 clinically normal subjects designated by different symbols, and the same symbol for each subject was used in three assays. All of them delivered during the 40th week, except FHM and SA, who delivered 1 week later.

The results of analysis of hPL, E_3, and E_4 of each patient with complicated pregnancy are given in Figures 5–8, along with clinical findings in the legends of each figure.

DISCUSSION

The nature of distribution curves, developed in our laboratory by analyzing 300 normal pregnancy sera for hPL and 460 sera for unconjugated E_3, are similar to those obtained by previous investigators.[19–22] The present data for hPL levels also confirm the concept that an abnormal value is one that falls below a certain limit. Although the absolute level of lower limits may vary slightly from laboratory to laboratory, we feel that the value lower than 4 μg/ml after 35 weeks of gestation is regarded as "fetal danger" zone. It is further observed that the hPL level diminishes due to postmaturity.

The unconjugated E_3 level in normal pregnancy increases at a certain rate between 30 and 34 weeks which then increases significantly after 35 weeks ("surge point"). This confirms the previous results obtained by Buster et al.[23,24] It has been further confirmed that the unconjugated E_3 level in normal pregnancy starts to diminish after 40 weeks.

The antisera, used to assay serum E_4 levels, was raised in rabbits by the use of E_4-6-conjugate antigen. This particular E_4 derivative is structurally sound to produce a specific antibody against E_4. This has been proved experimentally by the characteristics of the antisera which has been described in detail by one of us.[15] It is the most specific antisera raised so far.

The procedure to assay serum E_4 using this particular antisera has also been previously described.[15] It is simple and needs no prior chromatography for purification. The intra- and interassay variability are 7 and 14%, respectively. The sensitivity of the method has been improved to 25 pg/tube or 83 pg/ml serum by reducing the molarity of the buffer from 0.05M to 0.01M. The serum blank value is consistently indistinguishable from zero.

There are two methods for assay of serum E_4, described in the literature a) using the antisera raised against E_3-16,17-dihemisuccinate-BSA in sheep[4] and utilizing its high cross-reactivity for E_4; b) using the antisera raised against E_4-3-conjugate in rabbits.[12] These two methods were used to determine the serum E_4 levels in normal pregnancy by other investigators.[13,14,20]

The mean E_4 concentrations of serum from normal subjects in the third trimester obtained by our method

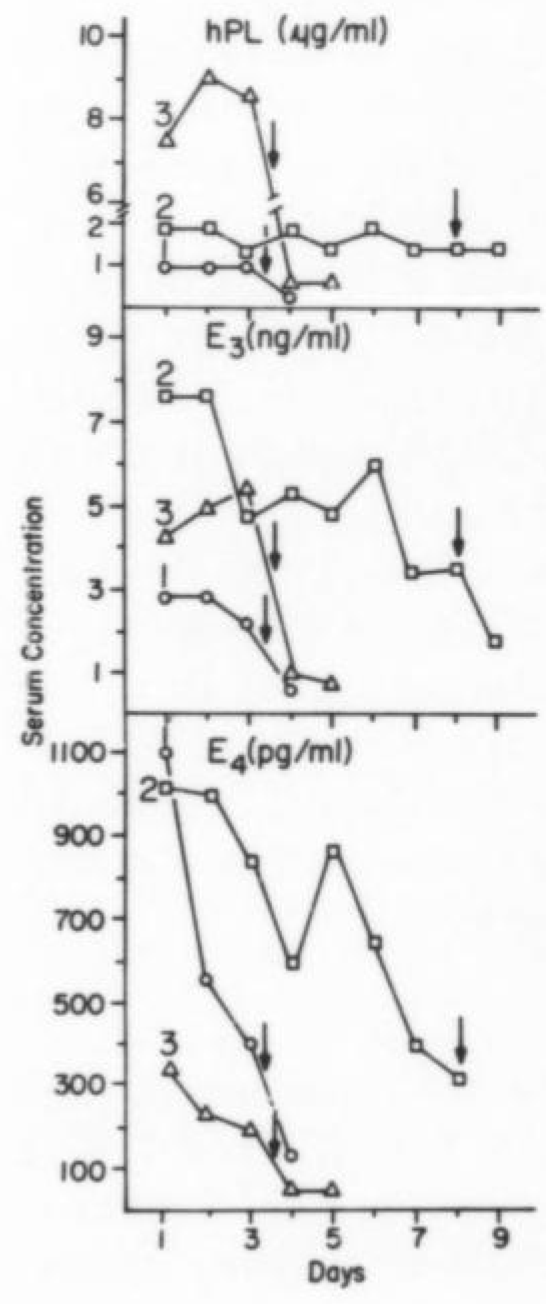

Fig 5. Daily serum E_4, and E_3, and hPL levels in 3 subjects designated by different symbols where intrauterine fetal deaths occurred, indicated by arrows.

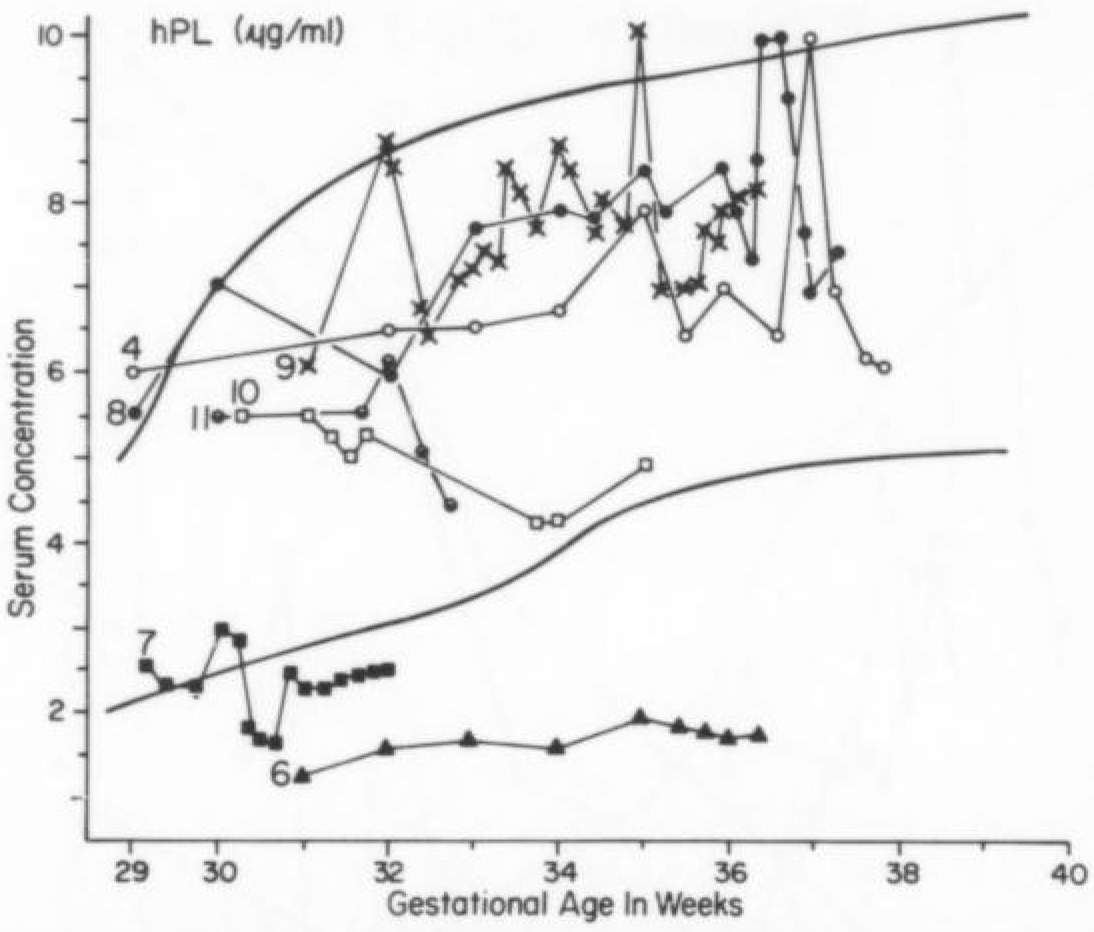

Fig 6. Serial serum hPL levels in 8 problem patients, designated by different symbols. (The same symbol for the same patient was used in Figures 7 and 8.) Clinical conditions and outcomes of these pregnancies are described in the following figures. Broad curved lines indicate the upper and lower normal limits.

(250–950 pg/ml) are slightly lower than those found by Tulchinsky et al[14] (300–1200 pg/ml) and much lower than those obtained by Giebenhain et al.[20] The lower values obtained in this procedure are logical since the antiserum used is more specific than those used by others.

An interesting observation has also been made both from normal distribution and serial individual values of E_4. A surge point for E_4, similar to that of E_3, also occurs between 35 and 36 weeks, which may be further explored to determine the exact date of gestation.[24] It is further observed that the E_4 level in normal pregnancy starts to diminish after 40 weeks similiar to hPL and E_3 levels.

We have an almost 50% American Indian population in our normal group of subjects studied. We are, however, unable to detect in them any significant difference of levels of these parameters from those of the general population.

In uncomplicated diabetic patients controlled with insulin, the parameters under consideration increased with gestational age as in normal pregnancy as shown in Cases 4 and 8 (Figures 7 and 8). The outcome of these pregnancies was normal and uneventful. But the results (Case 5) obtained in a patient with diabetes complicated with urinary tract infection were somewhat different (Figure 7). While there was a rise in the E_3 level, E_4 did not rise during this period of investigation. However, the outcome of this pregnancy was normal. The initial drop of the E_4 level during the early part of investigation might be due to some unexplainable reasons, but the pronounced decrease in both E_3 and E_4 during the last week of investigation was believed secondary to therapy with sulfisoxazole.

The observations on toxemic patients with or without diabetes were more interesting and enlightening. In 3 severely toxemic patients (Figure 5) the condition was manifested by maternal high blood pressure, proteinuria, and fetal distress which progressed to death prior to delivery. In 2 cases, the hPL levels were low and remained low throughout the study period, indicating chronic fetal distress, while in the third case, hPL values were unusually high possibly secondary to placental chorioangioma or increased placental mass. Thus, the daily hPL levels failed to signal acute fetal distress or predict intrauterine fetal death. Both serum E_3 and E_4 levels demonstrated a steady decrease in 2 cases during this period, but the E_4 concentration dropped dramatically at least 1 day earlier than E_3. In the third, while E_3 levels rose slightly, E_4 values started to drop. The initial abnormally high value of E_4 for that gestational age in 2 cases is unexplainable; it is definitely not due to an error of our method, as the value of quality control run simultaneously did not change. It is apparently a biologic vagary or it may be due to some unknown reasons.

In other cases, pregnancies were complicated by hypertension, proteinuria, diabetes, and/or increasing severity of maternal disease. The pregnancies terminated with the delivery of live, but seriously distressed babies, manifested by either late decelerations during OCT, low Apgar score at the time of cesarean section, or small-for-gestational-age babies. In all cases, the E_4 level decreased slightly or significantly (Case 6) while the E_3 level remained unchanged (Cases 6 and 11) or even increased slightly (Case 9) (Figures 7 and 8). Again, hPL levels did not reflect acute fetal distress.

The significant drop of E_3 in a patient (Case 6) during the early part of investigation is unexplainable; however, the E_4 level remained almost constant and the fetus survived (Figure 7). The pregnancy was terminated due to the patient's poor condition and late decelerations, and a small-for-gestational-age baby was delivered. In 1 patient (Case 9) the spikes found in all the measurements were not unusual in the sequential determinations of hormones (Figure 8). In another patient (Case 10), the drop of the E_4 level was more gradual, beginning from 32 weeks, while the E_3 level remained somewhat constant during that period, but started to drop after almost 1½ months (Figure 8). Possibly some unknown chronic distress to the fetus started from that time which was not signaled by E_3 value, but by the E_4 value. Patient BJ (Case 11) and Patient CP (Case 7) behaved similarly as Patient RI (Case 10) in respect to E_4 concentrations (Figures 7 and 8). Both values started to drop gradually, indicating a chronic fetal distress (Case 11) and severe

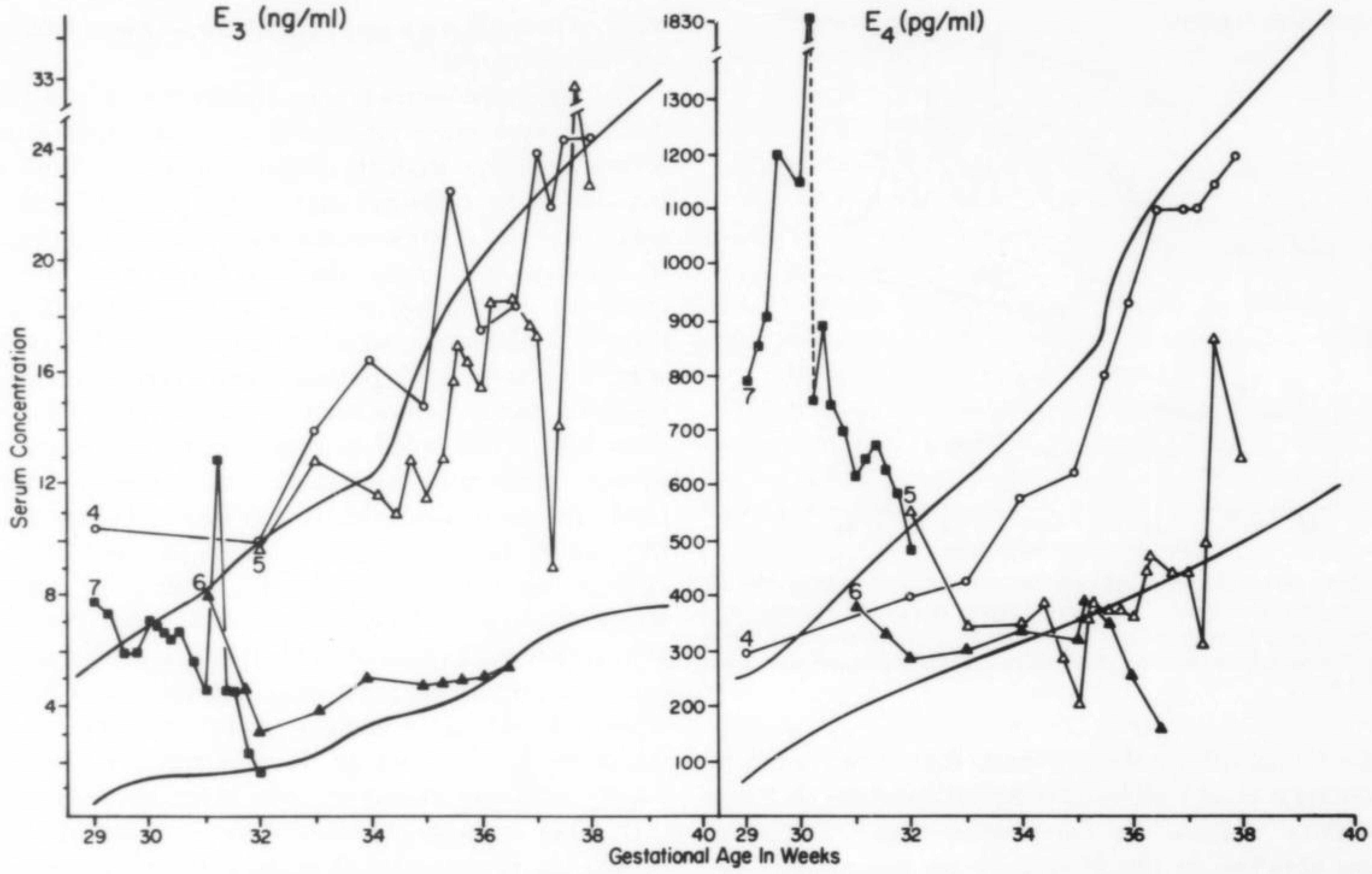

Fig 7. Serial serum E_3 and E_4 levels in 4 problem patients designated by different symbols as pointed out before. Broad curved lines indicate the upper and lower normal limits. (4) Uncomplicated diabetes Class B; (5) diabetes Class D, UTI; (6) essential hypertension, superimposed preeclampsia, diabetes Class D; (7) severe preeclampsia.

fetal distress (Case 7) (Figures 7 and 8). Due to the poor general condition of the patients, the pregnancies were terminated, and small-for-gestational-age babies were delivered, although the Apgar score and OCT appeared to be normal in Patient BJ (Case 11), but those for Patient CP (Case 7) were abnormal (Figures 7 and 8). The significance of the unusually high E_4 values at 31–32 weeks and at 30 weeks in the 2 patients (Cases 7 and 10) is at present not clear to us; again the corresponding E_3 values are also on the high side (Figures 7 and 8).

In one case (Case 7), we observed another interesting phenomenon (Figure 7): The E_4 level decreased dramatically when measured 8 hours after an OCT on the 30th week, and continued to diminish thereafter. The E_3 and hPL levels did not change. The significance of this fall of E_4 level is being further investigated. It is entirely possible that the OCT caused a borderline compromised fetus to become severely distressed.

One explanation of the metabolic events taking place is the following: There is considerable objective evidence that in chronic intrauterine growth retardation and in asphyxia the liver is not spared in chronic fetal distress. In acute fetal distress with fetal hypoxia, blood flow to the fetal liver, which is the site for E_4 synthesis, may be diminished. Hypoxia of liver cells would result in decreased E_4 production by limiting the 15α-hydroxylation. On the other hand, intrauterine fetal asphyxia and distress may increase the production of estriol precursor via the fetal pituitary–adrenal–stress response. Since the 16α-hydroxylation enzyme is also present in fetal adrenal. The E_3 synthesis is not seriously hampered.

Comparison and analysis of the data obtained from the simultaneous measurements of serum E_3, E_4, and hPL sequentially from these high-risk patients led us to hypothesize that the E_4 holds considerable promise to being an earlier and a more reliable indicator of fetal distress. The results of this limited number of patients, so far presented, are encouraging. A more comprehensive study along this line may lead to a conclusion that the determination of serum E_4 concentration may replace that of E_3 in the future for fetal assessment in high-risk pregnancies.

While this report was in preparation, Notation and Tagatz[25] published their observations about the importance of serum E_4 in complicated pregnancies. They concluded that serial concentrations of maternal serum E_4 did not appear to be clinically useful in the manage-

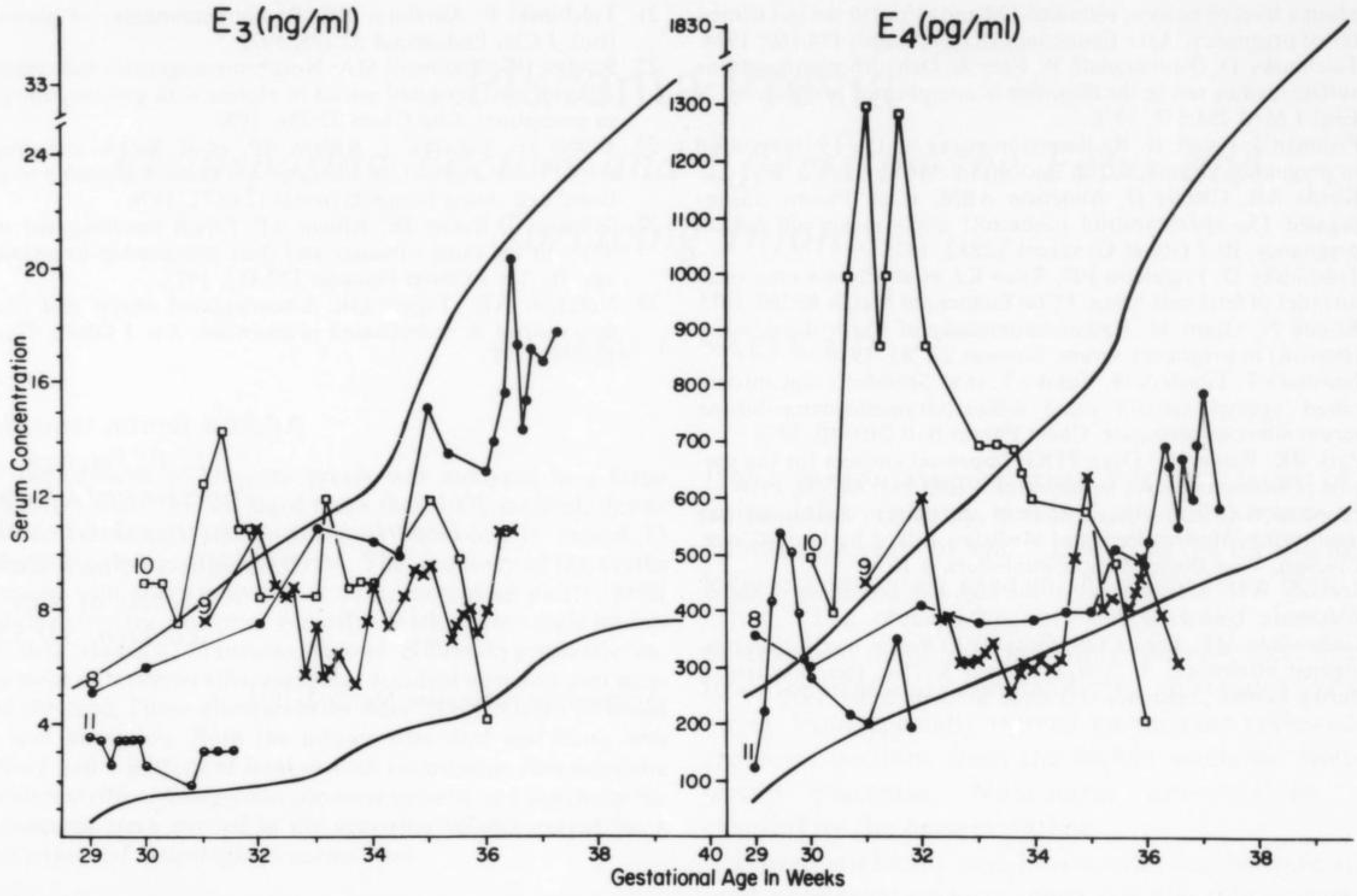

Fig 8. Serial serum E_3 and E_4 levels in 4 problem patients designated by different symbols as pointed out before. Broad curved lines indicate the upper and lower normal limits. (8) Uncomplicated diabetes Class D; (9) severe preeclampsia, complicated diabetes Class D; (10) severe preeclampsia; (11) essential hypertension, severe preeclampsia, diabetes Class D.

ment of high-risk pregnancies. While appreciating their statistical calculations for the interpretation of their data, we are, however, skeptical about their conclusion. First of all, their method for the assay of E_4 was based on the use of an antiserum, raised against an E_3 antigen which is, as reported by Gurpide et al,[4] not only nonspecific, but has a high variable plasma blank value. Second, each individual patient should be judged separately on the basis of E_3 and E_4 levels. It is not unusual to find some normal pregnant women starting with low levels of these estrogens that remain low, or increase slightly, throughout the pregnancy, which is normal. On the other hand, some start with high levels that remain high throughout the pregnancy. Therefore, it is our opinion that one should not interpret the values superimposed on the normal limits; instead one should look at the general trend of the values for consecutive days to ascertain whether they are constant, rising, or decreasing. A 30–40% drop of value from the previous 1 of 2 consecutive days, even though they remain in normal limits, should be considered as an ominous sign of fetal distress.

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